

OrbitText: A Hybrid Prediction System for Efficient and Effortless Text Entry in Virtual Reality

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ABSTRACT

Text entry in virtual reality (VR) is a common task, however, it is often hindered by low typing efficiency and physical fatigue. To address these challenges, we present OrbitText, a novel text entry system for VR environments. OrbitText features a dual-handed, minimal-motion 3D interface and integrates an innovative typing prediction model that combines large language models (LLMs) with n-gram models. Our design aims to minimize physical strain while enhancing typing efficiency. User studies demonstrate that OrbitText significantly outperforms mainstream virtual keyboards, achieving higher efficiency, reduced fatigue, and a lower perceived task load.

Index Terms: Human-centered computing → Human computer interaction → Interaction techniques → Text input.

1 INTRODUCTION

Text entry in virtual reality is a common task, however, mainstream virtual keyboards often lead to muscle fatigue and discomfort due to prolonged hand movements in the absence of physical support [1]. Additionally, directly adapting conventional keyboard layouts and recommendation algorithms to the VR environment may not align with users' expectations and device usage habits, resulting in decreased efficiency [2]. Moreover, mainstream recommendation algorithms typically use n-gram models, which excel at capturing local contextual relationships and making quick predictions but struggle with long-range coherence. In contrast, LLM-based methods effectively capture long-range semantic dependencies, though they are less accurate for local text predictions [3].

To address these challenges, we present OrbitText, a highly efficient VR typing system designed to reduce muscle fatigue and physical task load. OrbitText integrates an LLM-assisted hybrid text prediction model with a dual-handed 3D UI that aligns with users' natural habits and expectations. This design effectively addresses common issues such as fatigue and slow typing, which are typically associated with virtual keyboards.

2 METHODS

2.1 User Interface

OrbitText's UI features multi-layered circular menus, controlled via dual VR headset controllers, and includes a predictive display window positioned beneath the input field, as shown in Fig. 1(a). Users can navigate through all menu elements using the joysticks and trigger buttons on the controllers. The controller's data, *i.e.*, the depth,

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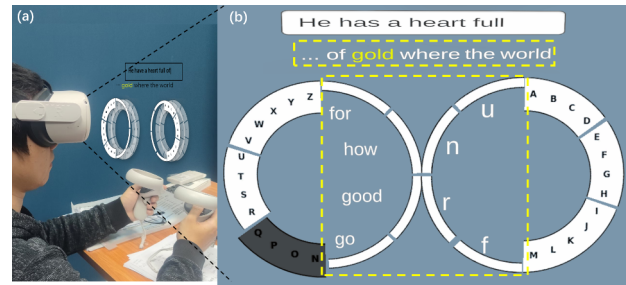


Figure 1: (a) User Interface of OrbitText. (b) The main keyboard layout, with highlighted regions for prediction-based text display, shows letter-level and word-level predictions within the circular menu, while sentence-level predictions are displayed underneath the input field.

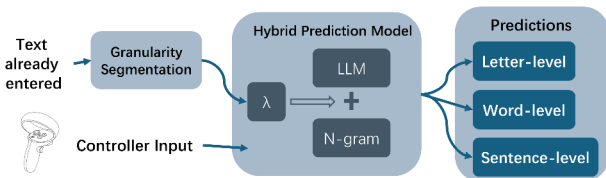


Figure 2: The flowchart of the prediction model

acceleration, and rotation, are used to navigate between menu layers and extract information from the prediction model.

Text input is utilized by a six-region keyboard layout, as shown in Fig. 1(b). Each region corresponds to a subset of letters. Rather than selecting individual letters, users tilt the joystick toward the desired region, enabling an “ambiguous input” mechanism similar to the T9 keyboard [4]. Recommended results from the prediction model are displayed in the highlighted area in Fig. 1(b), with each sub-region showing predictions at different levels of granularity.

2.2 Prediction Model

To improve text entry efficiency, we propose a hybrid prediction model that combines n-gram and large language models (LLMs). By adjusting the interpolation between the n-gram model and LLM using a hyperparameter λ , as shown in the following formula, the system optimizes the balance between local precision and global coherence, which improves typing speed and accuracy.

$$P(y|x) = \lambda P_{n\text{-gram}}(y|x) + (1 - \lambda) P_{LLM}(y|x) \quad (1)$$

The prediction model in OrbitText operates at three levels of granularity, with the hyperparameter dynamically adjusted to prioritize different prediction sources at each level, as illustrated in Fig. 2.

Letter-Level: N-gram predictions provide fast and accurate next-letter suggestions based on the preceding character.

Word-Level: Predictions combine spelling analysis, which evaluates typed letters to suggest completions, with semantic predictions from the hybrid model to propose contextually relevant words. The combined probabilities yield the four most likely word suggestions.

Sentence-Level: LLM-based predictions analyze preceding words to capture long-range dependencies and generate contextually coherent text, dynamically updating the visual interface.

To ensure a balance between precision and coherence, the weighting parameters are set as $\lambda_1 = 0.9$ for letter-level, $\lambda_2 = 0.5$ for word-level, and $\lambda_3 = 0.1$ for sentence-level predictions.

3 USER EXPERIMENTS

3.1 Experiment Design

User studies involved 15 unpaid participants (7 males, 8 females) from a local university, all right-handed with corrected vision to ensure compatibility with VR text entry tasks. Participants had varying VR experience levels and underwent a 20-phrase training session before completing three text entry tasks using all five keyboard configurations: a reference QWERTY keyboard and four OrbitText setups (no recommendations, n-gram only, LLM only, and hybrid). Task order was counterbalanced with a Latin square, and participants completed the NASA-TLX scale[5] after each task to evaluate workload.

The experiment used a Meta Quest 2 headset, with text for tasks randomly drawn from the Corpus of Contemporary American English (100 words per passage). OrbitText employed OpenAI's *text-embedding-ada-002* model and a 3-gram model based on the Google Books Ngram Corpus.

To evaluate the effectiveness of OrbitText in simplifying text entry tasks and improving efficiency, we introduced two new metrics: SK (Saved Keystrokes) and PTS (Percentage of Time Saved), defined by the following formula:

$$SK = \frac{Text_{total} - Text_{Orbit}}{Text_{total}} \quad PTS = \frac{Time_{ref} - Time_{Orbit}}{Time_{ref}} \quad (2)$$

SK represents the difference between the actual number of keystrokes and the number of word characters, indicating the assistance provided by the input method in terms of keystrokes. PTS represents the time saved using OrbitText compared to typing the same text on a traditional QWERTY keyboard in a VR environment.

3.2 Results

The SK and PTS results of the user experiment are shown in Fig. 3. Given that the measurement data follows a normal distribution, a repeated measures ANOVA was conducted to test the significance of the data, with results corrected using the Bonferroni method in SPSS 26.

A significant difference in keystroke counts was observed across methods ($F(3, 42) = 26.846, p < 0.001$). Post hoc tests showed the Hybrid method required fewer keystrokes than 'None' ($p < 0.001$), 'N-gram only' ($p < 0.001$), and 'LLM only' ($p < 0.05$). The 'LLM only' method also reduced keystrokes compared to 'None' ($p < 0.01$) and 'N-gram only' ($p < 0.05$), while 'N-gram only' did not differ significantly from 'None.'

For input time reduction, significant differences were also found ($F(3, 42) = 46.817, p = 0.000$). The Hybrid method outperformed 'None' ($p < 0.001$), 'N-gram only' ($p < 0.05$), and 'LLM only' ($p < 0.05$). The 'LLM only' method reduced input time more than 'None' ($p < 0.001$) but was inferior to 'N-gram only' ($p < 0.05$). 'N-gram only' significantly outperformed 'None' ($p < 0.001$).

The Hybrid model addressed the response delay of the 'LLM only' method, which impacted reaction times, effectively combining the strengths of both LLM and N-gram models for improved performance.

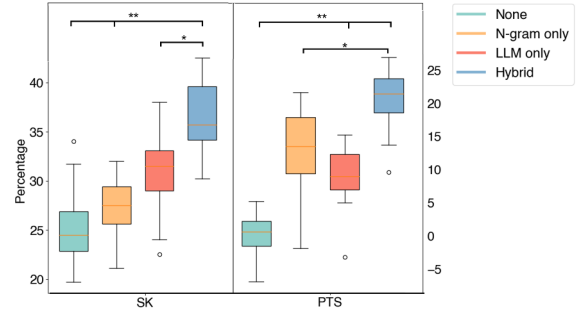


Figure 3: SK and PTS results of user experiment

The NASA-TLX results based on user assessments show that, compared to the QWERTY virtual keyboard in VR, OrbitText reduced Physical Demand (by 31.7%) and Temporal Demand (by 15.6%), but with higher Mental Demand (by 22.3%). Task load scores across different OrbitText configurations showed minimal variance.

4 DISCUSSION AND FUTURE WORK

User studies demonstrate that OrbitText, powered by the Hybrid model, effectively reduces keystrokes and enhances input efficiency. While it decreases physical and temporal workload, it may demand greater cognitive effort. Its ergonomic layout and context-aware predictions make it particularly suitable for VR tasks, especially for users with mobility impairments.

Future improvements focus on optimizing the UI to ease cognitive load and refining the hybrid model with better n-gram and LLM integration.

5 ACKNOWLEDGMENT

This work was supported by the National Key R&D Program of China (No.2022YFF0902303) and the 2022 major science and technology project "Yuelu-Multimodal Graph-Text-Sound-Semantic Gesture Big Model Research and Demonstration Application" in Changsha (No.kh2301019).

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